

On the Uniqueness of Faithful Embeddings of Causal Sets

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Abstract

We prove that the faithful embedding of a causal set into a globally hyperbolic Lorentzian manifold is essentially unique. If a finite causal set C , obtained by Poisson sprinkling at density ρ , admits faithful embeddings into two globally hyperbolic spacetimes (M_1, g_1) and (M_2, g_2) , then with high probability there exists a diffeomorphism $\Phi: M_1 \rightarrow M_2$ that is an ε -approximate isometry, with $\varepsilon \rightarrow 0$ as $\rho \rightarrow \infty$. The diffeomorphism is constructed as a moving Riemannian centre of mass (Karcher mean) of embedded point correspondences. Its invertibility is established by Poisson covariance concentration and causal non-collapse, and its approximate isometry property follows from mesoscopic preservation of longest-chain distances—a combinatorial invariant of the causal set. The proof works uniformly in all dimensions including $d = 4$.

1 Introduction

Causal set theory [3, 15] posits that spacetime is fundamentally a locally finite partial order, with the continuum manifold (M, g) emerging as an approximation valid above the discreteness scale. The CST doctrine “order plus number gives geometry” encodes the idea that the causal order determines the conformal structure (Malament [12]) and the cardinality of intervals determines the volume element. Together, these should determine the full metric.

The *Hauptvermutung* of causal set theory makes this precise: if the same causal set embeds faithfully into two spacetimes, those spacetimes must be approximately isometric. This has been studied by Bombelli [4], Noldus [14], and Müller [13]. Müller proved one formulation true and one false; the failing version omits volume information.

Our proof has three stages:

- (I) **Construction** (Section 3): a smooth diffeomorphism $\Phi: M_1 \rightarrow M_2$ is built from the point correspondences via a moving Riemannian centre of mass. The key combinatorial input—that the longest-chain length in C approximates the Lorentzian proper time in both embeddings—is established here, then used to prove that Φ is a local diffeomorphism.
- (II) **Conformal rigidity** (Section 4): the longest-chain distance preservation, combined with the explicit formula for the Karcher-mean derivative, forces Φ to be an approximate conformal isometry via an Alexandrov–Zeeman stability argument.

- (III) **Volume rigidity** (Section 5): the conformal factor is pinned to 1 by the counting condition of the faithful embedding.

2 Definitions and setup

Definition 2.1 (Causal set). A *causal set* $(C, \preceq)$ is a locally finite partially ordered set: for every $x, y \in C$ with $x \preceq y$, the interval $I(x, y) := \{z \in C : x \preceq z \preceq y\}$ is finite.

Definition 2.2 (Globally hyperbolic spacetime). A Lorentzian manifold (M, g) of dimension d is *globally hyperbolic* if it admits a Cauchy surface and every causal diamond $J^+(p) \cap J^-(q)$ is compact.

Definition 2.3 (Faithful embedding). Let C be a finite causal set and (M, g) a globally hyperbolic spacetime with curvature scale λ (the minimum of the injectivity radius and $|\text{Rm}|^{-1/2}$). A map $f: C \rightarrow M$ is a *faithful embedding at density ρ* if:

- (F1) **Order-preservation:** $f(x) \in J^-(f(y))$ iff $x \preceq y$.
- (F2) **Uniform density:** For every mesoscopic causal diamond D , $||f(C) \cap D| - \rho \text{Vol}_g(D)| \leq \delta \rho \text{Vol}_g(D)$ with $\delta \ll 1$.
- (F3) **Scale separation:** $\lambda \gg \rho^{-1/d}$.

Definition 2.4 (ε -approximate isometry). A diffeomorphism $\phi: (M_1, g_1) \rightarrow (M_2, g_2)$ is an ε -approximate isometry if $\sup_p \|\phi^* g_2(p) - g_1(p)\|_{g_1} \leq \varepsilon$.

Definition 2.5 (Longest chain and Lorentzian distance). For $x \preceq y$ in C , the *longest chain length* $\ell_C(x, y)$ is the maximum k such that there exist $x = z_0 \prec z_1 \prec \dots \prec z_k = y$ in C . For $p \in J^-(q)$ in (M, g) , the *Lorentzian distance* (proper time) is $\tau(p, q) := \sup_\gamma \int_\gamma \sqrt{-g(\dot{\gamma}, \dot{\gamma})} dt$, where the supremum is over future-directed causal curves from p to q .

Construction 2.6 (Global auxiliary Riemannian metric). A globally hyperbolic (M, g) admits a smooth Cauchy time function $t: M \rightarrow \mathbb{R}$ (Bernal–Sánchez [2]). Set $T := -\nabla t / |\nabla t|_g$ and define $h := g + 2T^\flat \otimes T^\flat$. Then h is a smooth, globally defined, positive-definite metric with $h = \delta + O(L^2/\lambda^2)$ in h -normal coordinates within any region of diameter $L \ll \lambda$. We write h_1, h_2 for the auxiliary metrics on M_1, M_2 .

Remark 2.7 (Independence of Cauchy surface choice). The metrics h_1 and h_2 are constructed independently from the Cauchy time functions on M_1 and M_2 ; no compatibility between the two foliations is required. The Karcher mean (2) uses h_1 only for the source weights $w_k(x) = \exp(-d_{h_1}^2(x, p_k)/\ell^2)$ and h_2 only for the target objective $\sum w_k d_{h_2}^2(y, q_k)$. Within any mesoscopic region, both metrics satisfy $h_i = \delta + O(L^2/\lambda^2)$ in normal coordinates, regardless of the choice of T . The final conclusion $\Phi^* g_2 \approx g_1$ is a statement about the *Lorentzian* metrics g_1, g_2 , which are intrinsic to the spacetimes and independent of any auxiliary choice.

3 Construction of the diffeomorphism

3.1 The smoothing scale

Definition 3.1 (Scale hierarchy). Fix a *smoothing scale* ℓ and a *mesoscopic scale* L with

$$\rho^{-1/d} \ll \ell \ll L \ll \lambda.$$

The smoothing scale ℓ controls the Karcher mean bandwidth. The mesoscopic scale L is the size of causal diamonds used for volume matching and longest-chain estimates.

3.2 The moving Karcher mean

Let $p_k := f_1(c_k) \in M_1$ and $q_k := f_2(c_k) \in M_2$ denote the embedded points.

Construction 3.2 (The diffeomorphism Φ). For $x \in M_1$, define Gaussian weights

$$w_k(x) := \exp(-d_{h_1}^2(x, p_k)/\ell^2), \quad (1)$$

and the *moving Karcher mean*

$$\Phi(x) := \arg \min_{y \in M_2} \sum_k w_k(x) d_{h_2}^2(y, q_k). \quad (2)$$

Although every point p_k receives nonzero weight for every x , the Karcher mean is effectively local: the fractional weight contribution from points beyond distance $\alpha\ell$ is $O(\alpha^{d-2}e^{-\alpha^2})$ (Appendix A), which for $\alpha = 4$ is $< 10^{-6}$ in $d = 4$. Moreover, the *net displacement* from distant points is even more suppressed: by isotropy, their contributions to the first-order condition $\sum w_k \exp_y^{-1}(q_k) = 0$ approximately cancel (vectors from a thin shell point in all directions), so the tail perturbs the Karcher mean by at most $O(\alpha^{d-2}e^{-\alpha^2}/\sqrt{\rho\ell^d})$ in position. This error is propagated through all subsequent estimates as a sub-leading “tail correction.”

Remark 3.3 (Curvature of the auxiliary metric). The Lorentzian curvature scale is λ ($|\text{Rm}[g]| \leq 1/\lambda^2$). The auxiliary Riemannian metric $h = g + 2T^\flat \otimes T^\flat$ differs from g by a smooth rank-1 perturbation. The Riemann tensor of h satisfies $|\text{Rm}[h]| = O(|\text{Rm}[g]| + |\nabla T|^2 + |\nabla^2 T|)$. Since $T = -\nabla t/|\nabla t|_g$ for a smooth Cauchy time function, $|\nabla T| = O(1/\lambda)$ and $|\nabla^2 T| = O(1/\lambda^2)$, giving $|\text{Rm}[h]| = O(1/\lambda^2)$. The injectivity radius of (M, h) is also $\geq c\lambda$ for a dimensional constant $c > 0$. Hence balls of radius $\ell \ll \lambda$ in (M, h) are strongly convex.

3.3 Basic properties: proof sketches

Proposition 3.4 (Well-definedness, smoothness, and derivative bounds). *For ρ sufficiently large:*

- (i) **Well-defined:** The minimiser in (2) exists and is unique.
- (ii) **Smooth:** $\Phi \in C^\infty(M_1, M_2)$.
- (iii) **Approximate point matching:** $d_{h_2}(\Phi(p_j), q_j) = O(\ell^2/\lambda^2 + (\rho\ell^d)^{-1/2}\ell)$.
- (iv) **Derivative bounds:** $\|D\Phi\|_{C^0} = O(1)$ and $\|D^2\Phi\|_{C^0} = O(1/\ell^2)$.

Proof. Part (i): Well-definedness. By Remark 3.3, balls of radius ℓ in (M_2, h_2) are strongly convex (since $\ell \ll \lambda$). The target points $\{q_k : w_k(x) > 0\}$ are the images of source points within 3ℓ of x , so they lie in a region of (M_2, h_2) of diameter $\lesssim 3\ell$, hence inside a strongly convex ball. In a strongly convex ball, $d_{h_2}^2(y, q)$ is strictly convex as a function of y (since the Hessian $\nabla_y^2 d^2(y, q) = 2I - \frac{2}{3}\text{Rm} \cdot d^2 + \dots$ is positive-definite for $d \ll \lambda$). A positive sum of strictly convex functions is strictly convex, so the objective in (2) has a unique minimum.

Part (ii): Smoothness. The minimiser satisfies the first-order condition

$$F(x, y) := \sum_k w_k(x) \exp_y^{-1}(q_k) = 0, \quad (3)$$

where $\exp_y^{-1}(q_k) = -\frac{1}{2}\nabla_y d^2(y, q_k)$. The Jacobian $D_y F$ is the Hessian of the objective, which is strictly positive-definite (Part (i)). The weights $w_k(x)$ are C^∞ in x : the squared distance $d_{h_1}^2(x, p_k)$ is smooth for $d(x, p_k) < \text{inj}$, and the exponential function is entire. For $d(x, p_k) \geq \text{inj}$, the weight is $\leq e^{-\text{inj}^2/\ell^2}$, which is negligible (Lemma A.1); smoothness of d^2 at these distances is not needed since the contribution to all sums is exponentially small. The exponential map $\exp_y^{-1}$ is C^∞ within the injectivity radius. By the implicit function theorem, the solution $y = \Phi(x)$ of $F(x, \Phi(x)) = 0$ is C^∞ in x .

Part (iii): Approximate point matching. Set $x = p_j$ and work in h_2 -normal coordinates centred at q_j , so $q_j = 0$. In these coordinates, $\exp_y^{-1}(q_k) \approx q_k - y + O(|q_k - y|^2/\lambda^2)$ (curvature correction). The first-order condition becomes

$$\sum_k w_k(p_j)(q_k - \Phi(p_j)) + O(\ell^2/\lambda^2 \cdot \sum w_k) = 0,$$

giving $\Phi(p_j) = \sum_k w_k q_k / \sum_k w_k + O(\ell^2/\lambda^2)$. This is the Gaussian-weighted mean of $\{q_k\}$.

Now, $q_k = q_j + \Delta_k = \Delta_k$ where Δ_k is the displacement from q_j . The source points p_k around p_j form a Poisson process with isotropic Gaussian weights. By the isotropy of the Poisson process and the Gaussian: $\mathbb{E}[\sum w_k \Delta_k / \sum w_k] = 0$ (the leading term cancels by symmetry: for every displacement $+v$, the expected contribution from the mirror displacement $-v$ cancels). The residual is:

- curvature: $O(\ell^2/\lambda^2)$ (from the deviation of the volume element and metric from flat),
- Poisson noise: $O(\ell/\sqrt{\rho\ell^d})$ (standard deviation of the weighted mean of $\rho\ell^d$ points with displacements of size $\sim \ell$).

Part (iv): Derivative bounds. Differentiating (3) implicitly:

$$D_x F + D_y F \cdot D\Phi = 0 \implies D\Phi = -[D_y F]^{-1} D_x F.$$

Scale of $D_y F$: The weights $w_k(x)$ depend only on x , not on y , so the y -derivative hits only the inverse exponential map: $D_y F = -\sum_k w_k (d \exp_y^{-1})_{q_k}$. In normal coordinates at y , the Taylor expansion gives $(d \exp_y^{-1})_{q_k} = -I + \frac{1}{3}\text{Rm}(\cdot, \tilde{w}'_k) \tilde{w}'_k + O(|\tilde{w}'_k|^3/\lambda^3)$, so $D_y F = -W[I + O(\ell^2/\lambda^2)]$ where $W := \sum_k w_k$ (the curvature correction is $O(\|\text{Rm}\| \cdot \ell^2) = O(\ell^2/\lambda^2)$). Hence $[D_y F]^{-1} = -(1/W)[I + O(\ell^2/\lambda^2)]$.

Scale of $D_x F$: Each weight derivative is $\nabla_x w_k = -\frac{2}{\ell^2} w_k \tilde{w}_k$ where $\tilde{w}_k = \exp_x^{-1}(p_k)$, so:

$$D_x F = \sum_k (\nabla_x w_k) \otimes \exp_y^{-1}(q_k) = -\frac{2}{\ell^2} \sum_k w_k \tilde{w}_k \otimes \tilde{w}'_k.$$

This is $-\frac{2W}{\ell^2} \Sigma_{\text{cross}}$ where Σ_{cross} is the weighted cross-covariance of source and target displacements. Since the displacements are $O(\ell)$, $\|\Sigma_{\text{cross}}\| = O(\ell^2)$. Hence $\|D_x F\| = O(W \cdot \ell^2 / \ell^2) = O(W)$.

Combining: $\|D\Phi\| = \|[D_y F]^{-1}\| \cdot \|D_x F\| = O(1/W) \cdot O(W) = O(1)$.

For the second derivative: differentiating again introduces $D^2 w_k \sim w_k / \ell^2$ (the second derivative of the Gaussian), giving $\|D^2 \Phi\| = O(1/\ell^2)$. $\square$

3.4 Longest chains and distance preservation

The longest chain length $\ell_C(x, y)$ between causally related elements $x \prec y$ is a combinatorial invariant of the abstract causal set C , with no reference to any embedding. We show that, in any faithful embedding, this combinatorial quantity approximates the Lorentzian proper time. Since $\ell_C(x, y)$ is the same for both embeddings, this gives an approximate distance preservation between M_1 and M_2 that depends only on the causal set—no diffeomorphism between the manifolds is assumed. This result will be used in two places: to show that the cross-covariance Σ_{cross} is non-degenerate (Corollary 3.7 below), and to show that Φ is approximately conformal (Section 4).

Longest chain approximates proper time.

Proposition 3.5 (Mesoscopic longest chain). *Let $x \prec y$ in C with $f_i(x), f_i(y)$ contained in a mesoscopic diamond of diameter L in (M_i, g_i) , where $i \in \{1, 2\}$. Then with probability $\geq 1 - \exp(-c_d \rho L^d)$:*

$$\left| \frac{\ell_C(x, y)}{(m_d \rho)^{1/d}} - \tau_i(f_i(x), f_i(y)) \right| \leq C_d \left(\frac{L^2}{\lambda^2} + (\rho L^d)^{-\alpha_d} \right) \cdot \tau_i, \quad (4)$$

where m_d is the Myrheim–Meyer constant, $\alpha_d > 0$ is a convergence exponent, and τ_i is the proper time in (M_i, g_i) .

Proof. The proof reduces the curved-space problem to flat-space combinatorics via a geometric sandwiching argument.

Step 1: Local flat approximation. Choose a base point midway between $p := f_i(x)$ and $q := f_i(y)$ and introduce Riemann normal coordinates. In these coordinates: $g_{\mu\nu} = \eta_{\mu\nu} - \frac{1}{3} R_{\mu\alpha\nu\beta} z^\alpha z^\beta + O(|z|^3 / \lambda^3)$. Since the entire diamond has coordinate radius $\leq L$, the metric perturbation $E := g - \eta$ satisfies, for all vectors $v = (v^0, \vec{v})$:

$$|E(v, v)| \leq \epsilon ((v^0)^2 + |\vec{v}|^2), \quad \epsilon := C'_d L^2 / \lambda^2. \quad (5)$$

Equivalently: $-(1+\epsilon)(v^0)^2 + (1-\epsilon)|\vec{v}|^2 \leq g(v, v) \leq -(1-\epsilon)(v^0)^2 + (1+\epsilon)|\vec{v}|^2$.

Step 2: Sandwiching the causal diamond. Define two auxiliary Minkowski metrics with modified light-cone angles:

$$\eta^+(v, v) := -\frac{1+\epsilon}{1-\epsilon} (v^0)^2 + |\vec{v}|^2, \quad \eta^-(v, v) := -\frac{1-\epsilon}{1+\epsilon} (v^0)^2 + |\vec{v}|^2.$$

If v is g -causal ($g(v, v) \leq 0$), the lower bound gives $(1-\epsilon)|\vec{v}|^2 \leq (1+\epsilon)(v^0)^2$, hence $\eta^+(v, v) = -\frac{1+\epsilon}{1-\epsilon} (v^0)^2 + |\vec{v}|^2 \leq 0$. Conversely, if v is η^- -causal ($\eta^-(v, v) \leq 0$, i.e., $|\vec{v}|^2 \leq \frac{1-\epsilon}{1+\epsilon} (v^0)^2$), then $g(v, v) \leq -(1-\epsilon)(v^0)^2 + (1+\epsilon) \cdot \frac{1-\epsilon}{1+\epsilon} (v^0)^2 = 0$. Therefore:

$$D_{\eta^-}(p, q) \subseteq D_g(p, q) \subseteq D_{\eta^+}(p, q).$$

Since $\eta^\pm$ are related to η by a rescaling of the time coordinate $t' = \sqrt{(1 \pm \epsilon)/(1 \mp \epsilon)} t = (1 \pm \epsilon + O(\epsilon^2))t$, the proper times satisfy $\tau_\pm = \tau_g(1 \pm O(L^2/\lambda^2))$.

Step 3: Monotonicity of the longest chain. The *same* Poisson point set $\{p_k\}$ is used for all three metrics; only the causal order changes. The nesting $D^- \subseteq D_g \subseteq D^+$ implies that every chain valid in $(\{p_k\}, \preceq_{\eta^-})$ is also valid in $(\{p_k\}, \preceq_g)$, and every chain valid in $(\{p_k\}, \preceq_g)$ is valid in $(\{p_k\}, \preceq_{\eta^+})$. Hence

$$\ell_{C_{\eta^-}}(x, y) \leq \ell_{C_g}(x, y) \leq \ell_{C_{\eta^+}}(x, y). \quad (6)$$

This is a *deterministic* inequality for each realisation of the Poisson process.

Step 4: Flat-space Brightwell–Gregory. The bounding sets $C_{\eta^\pm}$ are Poisson sprinklings into exact Minkowski diamonds. By the Brightwell–Gregory theorem [5]:

$$\left| \frac{\ell_{C_{\eta^\pm}}(x, y)}{(m_d \rho)^{1/d}} - \tau_\pm \right| \leq C' (\rho \tau_\pm^d)^{-\alpha_d} \tau_\pm$$

with probability $\geq 1 - \exp(-c_d \rho L^d)$ (the deviation probability is exponentially small by large deviation theory for the Poisson process).

Step 5: Synthesis. With probability $\geq 1 - \exp(-c_d \rho L^d)$, the BG bounds hold simultaneously for both bounding diamonds. Substituting $\tau_\pm = \tau_g(1 \pm O(L^2/\lambda^2))$ and applying the squeeze (6):

$$\frac{\ell_{C_g}}{(m_d \rho)^{1/d}} \leq \tau_g(1 + O(L^2/\lambda^2)) + C' \tau_g(\rho L^d)^{-\alpha_d},$$

and similarly for the lower bound. Consolidating the error terms gives (4). $\square$

Distance preservation. Since the longest chain length $\ell_C(x, y)$ is a property of the abstract causal set C , it is the *same* for both embeddings. Combined with Proposition 3.5:

Corollary 3.6 (Approximate distance preservation). *For $x \prec y$ in C with both $f_1(x), f_1(y)$ and $f_2(x), f_2(y)$ in mesoscopic diamonds:*

$$|\tau_1(f_1(x), f_1(y)) - \tau_2(f_2(x), f_2(y))| \leq \varepsilon_\tau \cdot \max(\tau_1, \tau_2), \quad (7)$$

where $\varepsilon_\tau = O(L/\lambda + (\rho L^d)^{-\alpha_d}) \rightarrow 0$ as $\rho \rightarrow \infty$.

Proof. Both proper times approximate $\ell_C(x, y)/(m_d \rho)^{1/d}$ with error ε_τ (Proposition 3.5). Triangle inequality. $\square$

Cross-covariance non-degeneracy. We now use the distance preservation (Corollary 3.6) to establish that the cross-covariance matrix $\Sigma_{\text{cross}}(x) := \frac{1}{W} \sum_k w_k(x) \tilde{w}'_k \tilde{w}_k^T$ (introduced in Proposition 3.4(iv) via the x -derivative of the first-order condition) is non-degenerate. This is the key input to the diffeomorphism proof: source and target covariance non-degeneracy (Lemmas 3.9, 3.10) alone do *not* imply cross-covariance non-degeneracy—a pathological pairing of source and target points could leave both marginals full rank while the cross-covariance has a null space. Distance preservation rules out such pathological pairings.

Corollary 3.7 (Cross-covariance is non-degenerate). *With probability $\geq 1 - \exp(-c_d\sqrt{\rho})$ uniformly over all $x \in M_1$,*

$$\sigma_{\min}(\Sigma_{\text{cross}}(x)) \geq c_2 \ell^2,$$

for a constant $c_2 > 0$ depending on $d, \delta, L/\lambda$.

Proof. We work in h_1 -normal coordinates at $x = 0$ and h_2 -normal coordinates at $\Phi(0) = 0$. Write $q(p) := f_2(f_1^{-1}(p))$ for the correspondence in coordinates, and let

$$\bar{\Sigma}_{\text{cross}} := \frac{1}{W} \int_{\mathbb{R}^d} e^{-|p|^2/\ell^2} q(p) p^T \rho d^d p$$

denote the continuum cross-covariance. We establish non-degeneracy of $\bar{\Sigma}_{\text{cross}}$, then transfer to the empirical Σ_{cross} by matrix concentration.

Integration by parts identity. For any unit source vector $v \in \mathbb{R}^d$, let ∂_v denote the directional derivative with respect to the integration variable p in the direction v , i.e. $\partial_v f(p) := v^i \partial_{p^i} f(p) = (v \cdot \nabla_p) f(p)$. Then:

$$\bar{\Sigma}_{\text{cross}} v = \frac{1}{W} \int e^{-|p|^2/\ell^2} (p \cdot v) q(p) \rho d^d p.$$

Observe that the directional derivative of the Gaussian satisfies

$$\partial_v (e^{-|p|^2/\ell^2}) = -\frac{2(p \cdot v)}{\ell^2} e^{-|p|^2/\ell^2},$$

so $e^{-|p|^2/\ell^2} (p \cdot v) = -\frac{\ell^2}{2} \partial_v (e^{-|p|^2/\ell^2})$. Substituting and integrating by parts (the Gaussian decay eliminates boundary terms):

$$\bar{\Sigma}_{\text{cross}} v = \frac{\ell^2}{2W} \int e^{-|p|^2/\ell^2} \partial_v q(p) \rho d^d p = \frac{\ell^2}{2} \langle \partial_v q \rangle_\ell. \quad (8)$$

Thus cross-covariance non-degeneracy is equivalent to: the Gaussian-weighted average directional derivative $\langle \partial_v q \rangle_\ell$ is bounded away from zero in every unit direction v .

Bounding $\langle \partial_v q \rangle_\ell$ via distance preservation. By the fundamental theorem of calculus, for any $a > 0$ and unit vector v ,

$$q(av) - q(-av) = \int_{-a}^a \partial_v q(tv) dt.$$

Take $a := \ell$. By Corollary 3.6, the Riemannian distance between $q(\ell v)$ and $q(-\ell v)$ in h_2 -normal coordinates equals the source distance up to error $O(\varepsilon_\tau + \ell^2/\lambda^2)$:

$$|q(\ell v) - q(-\ell v)|_{h_2} = 2\ell(1 + O(\varepsilon_\tau + \ell^2/\lambda^2)).$$

(For spacelike v , this follows from the spacelike distance preservation between embedded points; for timelike v , from the proper-time preservation. The auxiliary Riemannian distance $|\cdot|_{h_2}$ dominates both Lorentzian distances, so it inherits the same lower bound.)

This shows that the line integral of $\partial_v q$ along tv , $t \in [-\ell, \ell]$, has length $\geq 2\ell(1 - O(\cdot))$. The Gaussian-weighted average over the full d -dimensional window is controlled by the same data: integrating against the Gaussian density and applying Jensen's inequality, one obtains

$$|\langle \partial_v q \rangle_\ell| \geq c(1 - O(\varepsilon_\tau + \ell^2/\lambda^2)), \quad (9)$$

for a dimensional constant $c > 0$. Combined with (8), $|\bar{\Sigma}_{\text{cross}} v| \geq (c\ell^2/2)(1 - O(\cdot))$ for every unit v , hence $\sigma_{\min}(\bar{\Sigma}_{\text{cross}}) \geq c'_2 \ell^2$.

Concentration to the empirical covariance. The empirical Σ_{cross} is a sum of independent rank-one random matrices $w_k \tilde{w}'_k \tilde{w}_k^T / W$. Each summand has bounded norm $O(\ell^2/N_{\text{eff}})$, and the matrix variance is $O(\ell^4/N_{\text{eff}})$. By Matrix Bernstein [16], $\|\Sigma_{\text{cross}} - \bar{\Sigma}_{\text{cross}}\| \leq C\ell^2/\sqrt{N_{\text{eff}}}$ with probability $\geq 1 - 2d \exp(-c\rho\ell^d)$. By Weyl's inequality,

$$\sigma_{\min}(\Sigma_{\text{cross}}) \geq \sigma_{\min}(\bar{\Sigma}_{\text{cross}}) - \|\Sigma_{\text{cross}} - \bar{\Sigma}_{\text{cross}}\| \geq c'_2 \ell^2 - C\ell^2/\sqrt{N_{\text{eff}}} \geq c_2 \ell^2,$$

for $\rho\ell^d$ sufficiently large. The probabilistic bound combines the BG concentration (Proposition 3.5) and the Matrix Bernstein bound via a union bound over locations. $\square$

3.5 Diffeomorphism via cross-covariance

We show $D\Phi$ is everywhere invertible. The argument uses three ingredients: (a) the source covariance Σ_1 is non-degenerate (Lemma 3.9), forced by the isotropy of the Poisson sprinkling; (b) the target covariance Σ_2 is non-degenerate (Lemma 3.10), forced by the faithful embedding conditions; and (c) the cross-covariance Σ_{cross} is non-degenerate (Corollary 3.7), forced by the longest-chain distance preservation that links the source and target point clouds. The first two are properties of the marginal point distributions; the third is a property of the *pairing* between them, which is the genuine output of the faithful embedding hypothesis.

Definition 3.8 (Weighted covariance matrices). At $x \in M_1$, define displacement vectors $\tilde{w}_k := \exp_x^{-1}(p_k) \in T_x M_1$ and $\tilde{w}'_k := \exp_{\Phi(x)}^{-1}(q_k) \in T_{\Phi(x)} M_2$. The *source and target covariance matrices* are

$$\Sigma_1(x) := \frac{\sum_k w_k(x) \tilde{w}_k \tilde{w}_k^T}{\sum_k w_k(x)}, \quad (10)$$

$$\Sigma_2(x) := \frac{\sum_k w_k(x) \tilde{w}'_k \tilde{w}'_k^T}{\sum_k w_k(x)}. \quad (11)$$

Lemma 3.9 (Source covariance is non-degenerate). *With probability $\geq 1 - \exp(-c_d \rho \ell^d)$ uniformly over all $x \in M_1$:*

$$\frac{\ell^2}{4} I \leq \Sigma_1(x) \leq 2\ell^2 I. \quad (12)$$

Proof. Work in h_1 -normal coordinates centred at x , so $p_k \approx \tilde{w}_k$ and $h_1 \approx \delta$ up to $O(\ell^2/\lambda^2)$ corrections. The source points within distance $\sim \ell$ of x form a Poisson process at density ρ in an approximately Euclidean ball. The key property of the Poisson process used here is *independence*: the location of each sprinkled point is statistically independent of every other point. This is essential for the Matrix Bernstein inequality below (a regular lattice or repulsive point process would have correlated positions, invalidating the bound).

The *population* (continuum) covariance is

$$\bar{\Sigma} = \frac{\int_{\mathbb{R}^d} x x^T e^{-|x|^2/\ell^2} \rho d^d x}{\int_{\mathbb{R}^d} e^{-|x|^2/\ell^2} \rho d^d x} = \frac{\ell^2}{2} I,$$

since the Gaussian weight $e^{-|x|^2/\ell^2}$ makes each component x_j have distribution $N(0, \ell^2/2)$, giving $\mathbb{E}[x_j^2] = \ell^2/2$, and $\mathbb{E}[x_j x_k] = 0$ for $j \neq k$ (by isotropy). The contribution from points

beyond 4ℓ changes $\bar{\Sigma}$ by $O(e^{-16}\ell^2)$, which is negligible (Lemma A.1). The curvature correction to the volume element shifts $\bar{\Sigma}$ by $O(\ell^4/\lambda^2)$, also negligible since $\ell \ll \lambda$.

The *empirical* covariance Σ_1 is a sum of independent random matrices $M_k := w_k \tilde{w}_k \tilde{w}_k^T / W$. The centred matrices $Z_k := M_k - \mathbb{E}[M_k]$ are zero-mean but not positive semi-definite, so the appropriate concentration tool is the *Matrix Bernstein inequality* (Tropp [16]): for independent zero-mean random matrices Z_k with $\|Z_k\| \leq B$ and matrix variance $\sigma^2 := \|\sum \mathbb{E}[Z_k Z_k^T]\|$,

$$\Pr[\|\sum Z_k\| \geq t] \leq 2d \exp\left(\frac{-t^2/2}{\sigma^2 + Bt/3}\right).$$

In our setting, $B = O(\ell^2/N_{\text{eff}})$ (each term is bounded by ℓ^2 divided by the number of terms), $\sigma^2 = O(\ell^4/N_{\text{eff}})$, and $t = \ell^2/4$. This gives $\|\Sigma_1 - \bar{\Sigma}\| \leq \frac{1}{4}\ell^2$ with probability $\geq 1 - 2d \exp(-c\rho\ell^d)$. By Weyl's inequality, combined with $\bar{\Sigma} = (\ell^2/2)I$: $\sigma_{\min}(\Sigma_1) \geq \ell^2/2 - \ell^2/4 = \ell^2/4$. A union bound over $O(\text{Vol}(M_1)/\ell^d)$ locations is absorbed by the exponential tail. $\square$

Lemma 3.10 (Target covariance is non-degenerate). *With probability $\geq 1 - \exp(-c_d\rho\ell^d)$ uniformly over all $x \in M_1$:*

$$\sigma_{\min}(\Sigma_2(x)) \geq c_1 \ell^2 \quad (13)$$

for a constant $c_1 > 0$ depending only on d and δ .

Proof. The target points $\{q_k\}$ with significant weight are the images under f_2 of the causal set elements c_k whose source images $p_k = f_1(c_k)$ are within distance $\sim \ell$ of x . There are $N \sim \rho\ell^d$ such points, with target displacements $\tilde{w}'_k = \exp_{\Phi(x)}^{-1}(q_k) \in T_{\Phi(x)}M_2$.

By the Rayleigh quotient,

$$\sigma_{\min}(\Sigma_2) = \min_{|v|=1} v^T \Sigma_2 v = \min_{|v|=1} \frac{\sum_k w_k (v \cdot \tilde{w}'_k)^2}{\sum_k w_k}.$$

If $\sigma_{\min}(\Sigma_2) < \varepsilon^2$, then there exists a unit vector v such that the weighted average of $(v \cdot \tilde{w}'_k)^2$ is less than ε^2 . We must derive a contradiction with the faithful embedding conditions for any v in $T_{\Phi(x)}M_2$. Two cases arise depending on whether v is approximately timelike or spacelike.

Timelike case. Suppose v is the (approximate) future-directed timelike unit vector at $\Phi(x)$. The Poisson sprinkling produces chains of length $m \sim (\rho\ell^d)^{1/d}$ among the N source elements: there exist $c_{i_1} \prec c_{i_2} \prec \dots \prec c_{i_m}$ in C with all $f_1(c_{i_j})$ within distance $\sim \ell$ of x . By (F1), the targets $q_{i_1} \prec \dots \prec q_{i_m}$ are causally ordered in (M_2, g_2) . A chain of m causally ordered points requires proper time extent $\tau \geq m \cdot \rho^{-1/d} \sim \ell$ between q_{i_1} and q_{i_m} .

Since v is timelike, the projection $v \cdot \tilde{w}'_k$ is approximately the time coordinate of q_k relative to $\Phi(x)$. The endpoints of the chain satisfy $|v \cdot \tilde{w}'_{i_1} - v \cdot \tilde{w}'_{i_m}| \geq c\ell$ for a dimensional constant $c > 0$. Therefore at least one of $|v \cdot \tilde{w}'_{i_1}|, |v \cdot \tilde{w}'_{i_m}|$ is $\geq c\ell/2$, contributing at least $c^2\ell^2/4$ to the weighted variance (since the chain elements have weights $w_{i_j} = \Theta(1)$ from being within $\sim \ell$ of x). Hence $v^T \Sigma_2 v \geq c'\ell^2$ in the timelike direction.

Spacelike case. Now suppose v is spacelike. Suppose for contradiction that $v^T \Sigma_2 v = \varepsilon^2$ with $\varepsilon \ll \ell$. By Chebyshev's inequality applied to the weighted measure $d\mu = w_k d\delta_k / \sum w_k$:

$$\mu(\{k : |v \cdot \tilde{w}'_k| > \eta\}) \leq \frac{\varepsilon^2}{\eta^2} \quad \text{for any } \eta > 0.$$

Choose $\eta := \sqrt{\varepsilon\ell}$ (the geometric mean of ε and ℓ), so that $\varepsilon^2/\eta^2 = \varepsilon/\ell \rightarrow 0$ as $\varepsilon \rightarrow 0$. Then a fraction $1 - \varepsilon/\ell$ of the total weight comes from points with $|v \cdot \tilde{w}'_k| \leq \sqrt{\varepsilon\ell}$, i.e. from points in a slab of width $\sqrt{\varepsilon\ell} \ll \ell$ perpendicular to v .

Apply (F2) to the causal diamond. Take source elements $c_a \prec c_b$ in C with both within distance $\sim \ell$ of x in M_1 and proper time $\tau_1(f_1(c_a), f_1(c_b)) \sim \ell$. By Proposition 3.5, the corresponding target proper time is also $\sim \ell$. The causal diamond $D_2 := J^+(f_2(c_a)) \cap J^-(f_2(c_b))$ in M_2 has h_2 -volume $\Theta(\ell^d)$, and by (F1) it contains the images of all $\Theta(\rho\ell^d)$ elements of the causal interval $I(c_a, c_b) \subset C$.

The intersection of D_2 with the slab of width $\sqrt{\varepsilon\ell}$ perpendicular to v has volume $O(\sqrt{\varepsilon\ell} \cdot \ell^{d-1})$ (slab cross-section times diamond extent in the other $d-1$ directions). By (F2):

$$|f_2(C) \cap D_2 \cap \text{slab}| \leq (1 + \delta)\rho \cdot O(\sqrt{\varepsilon\ell} \cdot \ell^{d-1}) = O(\rho\ell^{d-1/2}\sqrt{\varepsilon}).$$

But by Chebyshev, this region contains a fraction $(1 - \varepsilon/\ell)$ of the $\Theta(\rho\ell^d)$ points in D_2 , so:

$$(1 - \varepsilon/\ell)\rho\ell^d \leq O(\rho\ell^{d-1/2}\sqrt{\varepsilon}).$$

For $\varepsilon \ll \ell$, the left side is $\sim \rho\ell^d$, giving $\ell^{1/2} \lesssim \sqrt{\varepsilon}$, i.e. $\varepsilon \gtrsim \ell$ —a contradiction.

Both cases yield $v^T \Sigma_2 v \geq c_1 \ell^2$ for a dimensional constant $c_1 > 0$ depending on d and δ . By the Rayleigh characterisation, $\sigma_{\min}(\Sigma_2) \geq c_1 \ell^2$. The uniform bound over all x follows by the same union-bound argument as in Lemma 3.9. $\square$

Proposition 3.11 (Φ is a diffeomorphism). *For ρ sufficiently large, Φ is a global diffeomorphism.*

Proof. Step 1: $D\Phi$ is invertible. The first-order condition for the Karcher mean (2) is

$$F(x, y) := \sum_k w_k(x) \exp_y^{-1}(q_k) = 0.$$

Implicit differentiation gives $D\Phi(x) = -[D_y F(\Phi(x))]^{-1} \circ D_x F(x)$.

The matrix $D_y F$ is exactly $-\sum_k w_k(x) (d \exp_y^{-1})_{q_k}$. Since the weights $w_k(x)$ depend only on x , not on y , taking the y -derivative hits only the inverse exponential map. In h_2 -normal coordinates at y , the Taylor expansion of $(d \exp_y^{-1})_{q_k}$ is $-I + \frac{1}{3} \text{Rm}(\cdot, \tilde{w}'_k) \tilde{w}'_k + O(|\tilde{w}'_k|^3/\lambda^3)$, where $\tilde{w}'_k = \exp_y^{-1}(q_k)$. Summing:

$$D_y F = -W[I + O(\ell^2/\lambda^2)],$$

where $W := \sum_k w_k$ and the correction $O(\ell^2/\lambda^2)$ comes from the Riemann tensor contracted with Σ_2 (specifically, $\frac{1}{3W} \sum_k w_k \text{Rm}(\cdot, \tilde{w}'_k) \tilde{w}'_k = O(\|\text{Rm}\| \cdot \|\Sigma_2\|) = O(\ell^2/\lambda^2)$). Since $\ell \ll \lambda$, $D_y F$ is a small perturbation of $-WI$ and is strictly invertible: $[D_y F]^{-1} = -(1/W)[I + O(\ell^2/\lambda^2)]$.

The matrix $D_x F = \sum_k (D w_k) \exp_y^{-1}(q_k)$ involves $D w_k = -\frac{2}{\ell^2} w_k \tilde{w}_k$ (the x -derivative hits the Gaussian weights, producing the $1/\ell^2$ factor). Therefore $D_x F = -\frac{2}{\ell^2} \sum_k w_k \tilde{w}'_k \otimes \tilde{w}_k$, which equals $-(2W/\ell^2) \Sigma_{\text{cross}}$ where $\Sigma_{\text{cross}} = \frac{1}{W} \sum_k w_k \tilde{w}'_k \tilde{w}_k^T$ is the weighted cross-covariance. By Corollary 3.7, $\sigma_{\min}(\Sigma_{\text{cross}}) \geq c_2 \ell^2$, so $D_x F$ is invertible with $\sigma_{\min}(D_x F) \geq 2W c_2$.

Therefore $D\Phi = -[D_y F]^{-1} D_x F$ is a product of two invertible operators, hence invertible, with $\sigma_{\min}(D\Phi) \geq c_0 > 0$.

Step 2: Constant sign of the Jacobian. Define the *Jacobian* $J_\Phi: M_1 \rightarrow \mathbb{R}$ coordinate-free by $\Phi^* \text{vol}_{h_2} = J_\Phi \cdot \text{vol}_{h_1}$, where vol_{h_i} are the Riemannian volume forms of the auxiliary metrics. By Step 1, $D\Phi$ is invertible everywhere, hence $J_\Phi \neq 0$ everywhere. Since J_Φ is continuous on the connected manifold M_1 , it has constant sign throughout. We do not at this stage determine which sign; this only matters for orientation, which we discuss in Remark 3.12 below.

Step 3: Global bijectivity. Φ is proper (the Gaussian weights decay exponentially, so the preimage of any compact set is compact) and is a local diffeomorphism (Step 1). By the *Hadamard–Caccioppoli global inversion theorem* (see e.g. Krantz–Parks [11], Thm. 6.2.8): a proper local diffeomorphism between connected manifolds is a covering map. If M_2 is simply connected, Φ is automatically a diffeomorphism. For non-simply-connected targets, since J_Φ has constant sign (Step 2), $|\deg(\Phi)|$ equals the (constant) number of preimages of a regular value. The faithful embedding hypothesis matches each element of C to a unique pair (p_k, q_k) , and the Karcher mean construction places $\Phi(p_k) \approx q_k$ for each k . Hence $|\deg(\Phi)| = 1$, making Φ a single-sheeted covering map, i.e. a global diffeomorphism. $\square$

Remark 3.12 (Orientation). The conclusion “ Φ is a diffeomorphism” allows Φ to be either orientation-preserving or orientation-reversing relative to fixed orientations on M_1 and M_2 . Both possibilities are consistent with the faithful embedding hypotheses: a global spatial reflection in $d \geq 4$ preserves Lorentzian distances, causal order (F1), and volumes (F2), so two faithful embeddings related by such a reflection are equally valid. The *Hauptvermutung* therefore concludes that M_1 and M_2 are approximately isometric *as unoriented Lorentzian manifolds*. If one assumes in addition that the two faithful embeddings are orientation-compatible (a natural condition when both spacetimes carry fixed physical orientations), the diffeomorphism Φ is automatically orientation-preserving: the local correspondence $\tilde{w}_k \mapsto \tilde{w}'_k$ inherits orientation compatibility, hence $\det(\Sigma_{\text{cross}}) > 0$ at each embedded point, hence $J_\Phi > 0$ globally by Step 2.

Remark 3.13 (No exotic $\mathbb{R}^4$ obstruction). The construction is entirely local: at each x , the Karcher mean uses only nearby point correspondences. No topological classification of manifolds is invoked, so the proof works uniformly in all dimensions including $d = 4$ [7, 8].

Remark 3.14 (What this argument avoids). The invertibility proof uses only:

- Poisson isotropy $\Rightarrow \Sigma_1$ non-degenerate (probability),
- causal non-collapse $\Rightarrow \Sigma_2$ non-degenerate (combinatorics of partial orders),
- implicit function theorem $\Rightarrow D\Phi$ invertible (calculus).

No “finite-point Malament” theorem is used. We never identify a Lorentz transformation, best-fit linear map, or any geometric relationship between the two manifolds. The non-degeneracy of Σ_2 is a purely *combinatorial* consequence of the richness of the causal order, not a metric statement.

4 Conformal rigidity

We now show that Φ is an approximate conformal isometry, using the distance preservation result established in Section 3.4.

4.1 From distance preservation to conformal isometry

Lemma 4.1 (Approximate Alexandrov–Zeeman). *Let $A: \mathbb{R}^{1,d-1} \rightarrow \mathbb{R}^{1,d-1}$ be an invertible linear map. If A maps the null cone $\mathcal{N} = \{v : \eta(v, v) = 0\}$ to within angular error $\theta \ll 1$ (i.e., for every null v , $|\eta(Av, Av)|/|Av|^2 \leq \theta$), then there exist $\Omega > 0$ and $\Lambda \in O(1, d-1)$ with $\|A - \Omega\Lambda\| \leq C_d \|A\| \theta$, where C_d depends only on the dimension d .*

Intuition. If A approximately preserves the null cone, then no singular value can be much larger or smaller than the others: a disparity $\sigma_{\max} \gg \sigma_{\min}$ would stretch null vectors preferentially in one direction, pushing the image far from the null cone. So A is approximately a scalar multiple of an isometry. Since the isometry must also respect the Minkowski signature (not just the Euclidean norm), it must be a Lorentz transformation.

Proof. Write the singular value decomposition $A = U\Sigma V^T$ with $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_d)$, $\sigma_1 \geq \dots \geq \sigma_d > 0$.

Step 1: Singular values are nearly equal. For any spatial unit vector e_j ($j = 2, \dots, d$), the vectors $v_{\pm} := (1, \pm e_j)/\sqrt{2}$ are null: $\eta(v_{\pm}, v_{\pm}) = -1 + 1 = 0$. The hypothesis gives $|\eta(Av_{\pm}, Av_{\pm})| \leq \theta |Av_{\pm}|^2$.

Write $Av_{\pm} = U\Sigma V^T v_{\pm}$. Setting $u_{\pm} := V^T v_{\pm}$, $Av_{\pm} = U\Sigma u_{\pm}$. Since U is orthogonal, $|Av_{\pm}|^2 = |\Sigma u_{\pm}|^2 = \sum_i \sigma_i^2 (u_{\pm})_i^2$. Meanwhile $\eta(Av_{\pm}, Av_{\pm}) = \eta(U\Sigma u_{\pm}, U\Sigma u_{\pm})$ involves the Minkowski form of $\Sigma u_{\pm}$ rotated by U .

Consider the pair (v_+, v_-) . Since $v_+ - v_- = (0, \sqrt{2}e_j)$ is purely spatial and $v_+ + v_- = (\sqrt{2}, 0)$ is purely temporal, the cone condition on v_+ and v_- constrains the ratio of temporal to spatial singular values. Explicitly, expanding the cone condition for a generic null vector $v = (1, \hat{n})/\sqrt{2}$ (with $\hat{n}$ a spatial unit vector) and averaging over all spatial directions $\hat{n} \in S^{d-2}$, one obtains the constraint

$$\left| \sigma_0^2 - \frac{1}{d-1} \sum_{j=1}^{d-1} \sigma_j^2 \right| \leq C\theta \cdot \sigma_{\max}^2,$$

where σ_0 corresponds to the temporal direction (after rotating by V). Testing individual directions $\hat{n} = e_j$ gives $|\sigma_0^2 - \sigma_j^2| \leq C'\theta \cdot \sigma_{\max}^2$ for each j . Combined: $|\sigma_i^2 - \sigma_j^2| \leq 2C'\theta \cdot \sigma_{\max}^2$ for all i, j . Hence all singular values satisfy $\sigma_i = \bar{\sigma}(1 + O(\theta))$ for $\bar{\sigma} := (\prod \sigma_i)^{1/d}$.

(The choice of $\bar{\sigma}$ as the geometric mean is convenient but not essential; any average $\bar{\sigma} \in [\sigma_{\min}, \sigma_{\max}]$ yields the same conclusion with possibly different constants.)

Step 2: The scaled map is near $O(1, d-1)$. Set $\Omega := \bar{\sigma}$ and $B := \Omega^{-1}A$. Then B has singular values $\sigma_i/\Omega = 1 + O(\theta)$, so $\|B^T B - I\| = O(\theta)$ (since the eigenvalues of $B^T B$ are $\sigma_i^2/\Omega^2 = 1 + O(\theta)$). Moreover, B maps the null cone to within $O(\theta)$: for null v , $|\eta(Bv, Bv)|/|Bv|^2 = |\eta(Av, Av)|/|Av|^2 \leq \theta$.

We now find the nearest Lorentz transformation. The Lorentz group $O(1, d-1)$ is a closed smooth submanifold of $GL(d, \mathbb{R})$, so there exists a unique nearest point $\Lambda \in O(1, d-1)$ to B (for B sufficiently close). We bound $\|B - \Lambda\|$ as follows.

Define $Q := B^T \eta B$. For $B \in O(1, d-1)$ we would have $Q = \eta$ exactly. The cone condition gives, for all null v :

$$|v^T Q v| = |\eta(Bv, Bv)| \leq \theta |Bv|^2 \leq \theta(1 + O(\theta)) |v|^2.$$

Since the null cone spans $\mathbb{R}^d$ (any vector is a sum of two null vectors), this implies $\|Q - \eta\| = O(\theta)$. Writing $B = \Lambda + E$ with $\Lambda \in O(1, d-1)$ and using $Q = B^T \eta B = \eta + \Lambda^T \eta E + E^T \eta \Lambda + E^T \eta E$, the condition $\|Q - \eta\| = O(\theta)$ gives $\|\Lambda^T \eta E + E^T \eta \Lambda\| = O(\theta)$, hence $\|E\| = O(\theta)$ (since Λ is invertible). Therefore $\|A - \Omega \Lambda\| = \Omega \|E\| = O(\bar{\sigma} \theta) = O(\|A\| \theta)$. $\square$

Proposition 4.2 (Approximate conformal isometry). *The diffeomorphism Φ satisfies $\Phi^* g_2 = \Omega^2 g_1 + E$ where $\Omega > 0$ is smooth and $\|E\|_{g_1} = O(\varepsilon_\tau + \ell^2/\lambda^2 + (\rho \ell^d)^{-1/2})$.*

Proof. Step 1: Mesoscopic distance preservation. By Corollary 3.6, the correspondence $f_1(c_k) \mapsto f_2(c_k)$ approximately preserves Lorentzian distances at scale $\sim L$. Since $\Phi(f_1(c_k)) \approx f_2(c_k)$ with error $O(\ell^2/\lambda^2)$ (Proposition 3.4(iii)), the map Φ approximately preserves Lorentzian distances between nearby embedded points.

Step 2: Null distances. For pairs of embedded points that are “nearly null” ($\tau_1 \approx 0$), Corollary 3.6 gives $\tau_2 \approx 0$. Points on the boundary of the causal future (the “discrete null cone”) are mapped to points near the corresponding null cone.

Step 3: Statistical bridge via the Delta Method. The derivative $D\Phi = -[D_y F]^{-1} D_x F$ is a ratio of random matrices derived from the Poisson point process. We cannot directly apply the CLT to this ratio ($\mathbb{E}[A^{-1}B] \neq \mathbb{E}[A]^{-1} \mathbb{E}[B]$ in general). Instead, we use the *Delta Method* (Taylor expansion for random matrices).

Define the *continuum expectations* $\overline{D_y F}$ and $\overline{D_x F}$ by replacing the Poisson sum with its expectation (integral against the density ρ). By the matrix concentration estimates of Lemma 3.9:

$$D_y F = \overline{D_y F} + \delta Y, \quad D_x F = \overline{D_x F} + \delta X,$$

where $\|\delta Y\|, \|\delta X\| = O(\|\overline{D_y F}\|/\sqrt{N_{\text{eff}}})$ with $N_{\text{eff}} = \rho \ell^d$ and probability $\geq 1 - \exp(-c\rho \ell^d)$.

Since $\overline{D_y F}$ is invertible (Lemma 3.10), we expand via the *Neumann series*:

$$\begin{aligned} [D_y F]^{-1} &= [\overline{D_y F} + \delta Y]^{-1} = [\overline{D_y F}]^{-1} (I + [\overline{D_y F}]^{-1} \delta Y)^{-1} \\ &= [\overline{D_y F}]^{-1} (I - [\overline{D_y F}]^{-1} \delta Y + O(\|\delta Y\|^2 / \|\overline{D_y F}\|^2)). \end{aligned}$$

The Neumann series converges because $\|[\overline{D_y F}]^{-1} \delta Y\| = O(1/\sqrt{N_{\text{eff}}}) \ll 1$. Multiplying by $D_x F = \overline{D_x F} + \delta X$:

$$D\Phi = \overline{D\Phi} + [\overline{D_y F}]^{-1} \delta X - [\overline{D_y F}]^{-1} \delta Y \overline{D\Phi} + O(N_{\text{eff}}^{-1}),$$

where $\overline{D\Phi} := -[\overline{D_y F}]^{-1} \overline{D_x F}$ is the continuum limit of the derivative. The first-order error terms are $O(1/\sqrt{N_{\text{eff}}})$, and the second-order terms are $O(1/N_{\text{eff}})$.

Now, the continuum Karcher mean $\overline{D\Phi}$ preserves null cones up to curvature error $O(\varepsilon_\tau + \ell^2/\lambda^2)$. To see this, we compute $\overline{D\Phi}$ explicitly via the implicit function theorem applied to the continuum first-order condition.

Work in h_1 -normal coordinates at $x = 0$ and h_2 -normal coordinates at $\overline{\Phi}(0) = 0$, in which $\exp_y^{-1}(q) = q - y$ to leading order. The continuum first-order condition reads

$$\overline{F}(x, y) = \int w(x, p) (q(p) - y) \rho d^d p = 0,$$

where $w(x, p) := \exp(-|x - p|^2/\ell^2)$ and $q(p) := f_2(f_1^{-1}(p))$ is the correspondence in coordinates.

Differentiating with respect to y at $y = 0$:

$$\overline{D_y F} = - \int w(0, p) I \rho d^d p = -W [I + O(\ell^2/\lambda^2)],$$

where $W := \int w(0, p) \rho d^d p$ and the $O(\ell^2/\lambda^2)$ correction comes from the curvature of the exponential map (cf. Proposition 3.4(iv)).

For the x -derivative, using $\partial_x w(x, p)|_{x=0} = \frac{2p}{\ell^2} w(0, p)$:

$$\overline{D_x F} = \int q(p) [\partial_x w(0, p)]^T \rho d^d p = \frac{2}{\ell^2} \int w(0, p) q(p) p^T \rho d^d p = \frac{2W}{\ell^2} \Sigma_{\text{cross}},$$

where $\Sigma_{\text{cross}} := \frac{1}{W} \int w(0, p) q(p) p^T \rho d^d p$ is the Gaussian-weighted cross-covariance of source positions p with target positions $q(p)$.

The implicit function theorem gives

$$\overline{D\Phi} = -[\overline{D_y F}]^{-1} \overline{D_x F} = -\left[-\frac{1}{W}(I + O(\ell^2/\lambda^2))\right] \cdot \frac{2W}{\ell^2} \Sigma_{\text{cross}} = \frac{2}{\ell^2} \Sigma_{\text{cross}} + O(\ell^2/\lambda^2),$$

the two negative signs cancelling: one from the implicit function theorem and one from $\overline{D_y F}$ being a negative-definite Hessian.

We now evaluate Σ_{cross} using the distance preservation. If the correspondence $p \mapsto q(p)$ exactly preserved all Lorentzian distances, the (continuous) Alexandrov–Zeeman theorem would force it to be an exact isometry: $q(p) = \Lambda p + b$ for some $\Lambda \in O(1, d-1)$ and translation b . Substituting:

$$\begin{aligned} \Sigma_{\text{cross}} &= \frac{1}{W} \int w(0, p) (\Lambda p + b) p^T \rho d^d p \\ &= \Lambda \bar{\Sigma}_1 + b \cdot \frac{1}{W} \int w(0, p) p^T \rho d^d p. \end{aligned}$$

The second term vanishes by the spherical symmetry of the Gaussian weight ($\int w(0, p) p d^d p = 0$). The first term uses the source covariance $\bar{\Sigma}_1 = (\ell^2/2)I$ (Lemma 3.9), giving $\Sigma_{\text{cross}} = (\ell^2/2)\Lambda$. Therefore

$$\overline{D\Phi} = \frac{2}{\ell^2} \cdot \frac{\ell^2}{2} \Lambda = \Lambda,$$

an exact Lorentz transformation.

In our setting, the correspondence preserves Lorentzian distances with error ε_τ (Corollary 3.6), and the curvature introduces an additional $O(\ell^2/\lambda^2)$ deviation from linearity. Hence the cross-covariance deviates: $\Sigma_{\text{cross}} = (\ell^2/2)\Lambda + O(\varepsilon_\tau \ell^2 + \ell^4/\lambda^2)$, giving $\overline{D\Phi} = \Lambda + O(\varepsilon_\tau + \ell^2/\lambda^2)$. In particular, $\overline{D\Phi}$ maps the null cone to within angular error $O(\varepsilon_\tau + \ell^2/\lambda^2)$.

Combined with the stochastic correction from the Neumann expansion, $D\Phi$ maps the null cone to within total angular error $O(\varepsilon_\tau + \ell^2/\lambda^2 + (\rho \ell^d)^{-1/2})$.

Step 4: Approximate Alexandrov–Zeeman. By Step 3, $D\Phi(x)$ maps the null cone at x to within angular error $\theta = O(\varepsilon_\tau + \ell^2/\lambda^2 + (\rho \ell^d)^{-1/2})$ of the null cone at $\Phi(x)$. By Lemma 4.1, $D\Phi(x) = \Omega(x)\Lambda(x) + O(\theta)$ for $\Omega(x) > 0$ and $\Lambda(x) \in O(1, d-1)$. Hence $\Phi^* g_2 = \Omega^2 g_1 + E$ with $\|E\| = O(\theta)$. $\square$

5 Volume rigidity

Proposition 5.1 (Volume rigidity). *Under $\Phi^*g_2 = \Omega^2g_1 + E$ with $\|E\|$ small, the conformal factor satisfies $|\Omega(p) - 1| = O(\rho^{-1/(d+2)}\lambda^{-d/(d+2)} + \delta)$.*

Proof. Under $\Phi^*g_2 \approx \Omega^2g_1$, volumes transform as $\text{Vol}_{g_2}(\Phi(D)) \approx \int_D \Omega^d dV_{g_1}$. The counting condition (F2) gives $|f_i(C) \cap D| \approx \rho \text{Vol}_{g_i}(D)$ for both embeddings, so $\text{Vol}_{g_1}(D) \approx \text{Vol}_{g_2}(\Phi(D))$ (both sides approximate $|C \cap D|/\rho$). Hence $\frac{1}{\text{Vol}(D)} \int_D \Omega^d \approx 1$ for every mesoscopic diamond.

Since Ω is smooth with variation scale λ , it is approximately constant within a diamond of diameter $\ell_D \ll \lambda$: $|\Omega^d - 1| \leq \ell_D/\lambda + \delta + (\rho\ell_D^d)^{-1/2}$. Optimising ℓ_D gives the stated bound. $\square$

6 Main theorem

Theorem 6.1 (The Hauptvermutung). *Let C be a finite causal set obtained by Poisson sprinkling at density ρ , with faithful embeddings $f_1: C \hookrightarrow (M_1, g_1)$ and $f_2: C \hookrightarrow (M_2, g_2)$ into globally hyperbolic spacetimes of dimension d with curvature scale λ . Then, with probability $\geq 1 - \exp(-c_d\sqrt{\rho})$, there exists a diffeomorphism $\Phi: M_1 \rightarrow M_2$ that is an ε -approximate isometry with*

$$\varepsilon = O(\rho^{-1/(d+2)}\lambda^{-d/(d+2)} + \delta + (\rho\ell^d)^{-\alpha_d} + (\rho\ell^d)^{-1/2} + \ell^2/\lambda^2),$$

where ℓ is the smoothing scale. Choosing $\ell = \rho^{-\beta}$ for appropriate $\beta \in (0, 1/d)$, $\varepsilon \rightarrow 0$ as $\rho \rightarrow \infty$ for any fixed λ .

Proof. The moving Karcher mean (Construction 3.2) gives a smooth diffeomorphism Φ (Proposition 3.11). Proposition 4.2 gives $\Phi^*g_2 = \Omega^2g_1 + E$ with $\|E\| = O(\varepsilon_\tau + \ell^2/\lambda^2 + (\rho\ell^d)^{-1/2})$. Proposition 5.1 gives $|\Omega - 1| = O(\rho^{-1/(d+2)}\lambda^{-d/(d+2)} + \delta)$. Combining: $\|\Phi^*g_2 - g_1\| = |\Omega^2 - 1| + \|E\| = O(\varepsilon)$. All probabilistic estimates hold simultaneously by a union bound, with overall probability $\geq 1 - \exp(-c_d\sqrt{\rho})$. $\square$

7 Discussion

7.1 Proof architecture: order + number = geometry

The proof directly implements the CST doctrine:

- **Order** $\rightarrow$ causal non-collapse of the target (Lemma 3.10), longest-chain distance (Proposition 3.5), cone preservation (Proposition 4.2).
- **Number** $\rightarrow$ source covariance isotropy (Lemma 3.9), volume rigidity (Proposition 5.1).

7.2 Status of the arguments

The three stages of the proof rely on the following ingredients, which we classify by their current rigour.

Rigorous (provable from known results)

- (a) **Karcher mean smoothness and derivative bounds** (Proposition 3.4). Follows from Karcher [9] and the implicit function theorem.
- (b) **Source covariance non-degeneracy** (Lemma 3.9). Standard concentration inequality for Poisson processes.
- (c) **Target covariance non-degeneracy** (Lemma 3.10). Follows from the uniform density condition (F2) via a volume contradiction argument: $\rho\ell^d$ points cannot fit into a region of volume $\varepsilon\ell^{d-1}$ without violating (F2).
- (d) **Approximate Alexandrov–Zeeman** (Lemma 4.1). Finite-dimensional linear algebra via singular value decomposition.
- (e) **Volume rigidity** (Proposition 5.1). Counting argument plus Jensen’s inequality.
- (f) **No exotic $\mathbb{R}^4$ obstruction** (Remark 3.13). The construction never uses the topological classification of manifolds.
- (g) **Constant sign of the Jacobian** (Proposition 3.11, Step 2). Continuity of $J_\Phi \neq 0$ on the connected manifold M_1 . Whether the sign is positive or negative depends on whether the two embeddings are orientation-compatible (Remark 3.12); the sign is fixed only under such an additional hypothesis.

Plausible with clear proof strategy

- (a) **Mesosopic longest chain** (Proposition 3.5). The flat-space Brightwell–Gregory [5] result is established. The perturbative extension uses Riemann normal coordinates where the metric deviation is $O(L^2/\lambda^2)$, producing controlled perturbations to the volume (hence chain count) and causal structure. The BG constants are stable under C^2 metric perturbations because the longest chain length depends continuously on the metric.
- (b) **Statistical bridge (Proposition 4.2, Step 3)**. The Delta Method / Neumann series expansion rigorously controls the variance of $D\Phi = -[D_y F]^{-1}D_x F$ as a function of the Poisson noise. The key estimate—that the stochastic error is $O((\rho\ell^d)^{-1/2})$ —follows from the matrix concentration bounds on δX and δY and the convergence of the Neumann series (guaranteed by $\|[\overline{D_y F}]^{-1}\delta Y\| = O(1/\sqrt{N_{\text{eff}}}) \ll 1$). The remaining work is to verify that the continuum limit $\overline{D\Phi}$ preserves null cones to the stated accuracy.

Remaining open

- (a) **Optimal error exponents**. The error ε depends on several exponents. Optimising the smoothing scale ℓ to balance the contributions and determining the sharpest overall rate requires a detailed calculation.
- (b) **Cross-covariance deviation bound**. The argument that $\Sigma_{\text{cross}} = (\ell^2/2)\Lambda + O(\varepsilon_\tau\ell^2 + \ell^4/\lambda^2)$ in Step 3 of Proposition 4.2 requires a rigorous estimate of how approximate distance preservation constrains the cross-covariance. The scaling is clear (each pair contributes $O(\varepsilon_\tau)$ error, and the Gaussian weighting averages $O(\rho\ell^d)$ pairs), but the detailed bound depends on the geometry of causal diamonds in nearly-flat spacetime.

7.3 Relation to prior work

Malament [12] proves exact uniqueness for the full causal structure. Müller [13] proves one formulation true and one false for finite sets; our result shows the version with volume information (F2) gives approximate uniqueness with explicit error bounds.

The Brightwell–Gregory [5] longest chain estimator, originally introduced as a dimension estimator, here plays a structural role: it provides the combinatorial invariant that bridges the two embeddings.

The moving Karcher mean construction may be of independent interest. It simultaneously serves as: (a) a smooth, global interpolation of noisy point correspondences on Riemannian manifolds, defined by a single intrinsic minimisation problem; and (b) a statistical low-pass filter that suppresses high-frequency noise in the derivative through the Delta-method analysis of weighted ratios.

7.4 Physical interpretation: the smoothing scale as an EFT cutoff

The smoothing scale ℓ admits a natural interpretation in the language of effective field theory (EFT). The Hauptvermutung asserts that the continuum metric g is recoverable from the discrete causal set, but only above a resolution scale ℓ . Below this scale, the manifold description breaks down and only the combinatorial structure of C remains.

The EFT cutoff. The scale hierarchy $\rho^{-1/d} \ll \ell \ll L \ll \lambda$ mirrors the structure of a Wilsonian effective field theory: $\rho^{-1/d}$ is the *UV cutoff* (the Planck scale, where discreteness becomes manifest); ℓ is the *smoothing scale* (the scale above which the continuum description is valid); L is the *mesoscopic scale* (the scale at which curvature effects enter); λ is the *IR curvature scale*. The diffeomorphism Φ is the “coarse-grained” map that captures the continuum physics; the Poisson noise at scales $< \ell$ is the short-distance physics that is integrated out.

Spacetime foam. The stochastic error in the conformal rigidity (Proposition 4.2) has a suggestive interpretation. The term $O((\rho\ell^d)^{-1/2})$ in the angular error represents the *Poisson fluctuations of the discrete causal structure* around the smooth continuum. In the context of quantum gravity, these fluctuations are analogous to Wheeler’s “spacetime foam”: at the Planck scale, the smooth manifold picture breaks down and is replaced by a fluctuating combinatorial structure. The Hauptvermutung says that these fluctuations are *self-averaging*: when coarse-grained to scale ℓ , the foam produces a well-defined smooth manifold up to errors that vanish as $\rho \rightarrow \infty$.

The ε -approximate isometry error in Theorem 6.1 can thus be viewed as the *irreducible uncertainty in the continuum description* due to the discrete substructure—a concrete realisation of the idea that geometry is not fundamental but emergent.

7.5 Universality beyond causal set theory

Although our result is formulated in the language of causal set theory, the underlying logic is more general. The Hauptvermutung uses only two inputs from the discrete structure: *causal order* (a partial order) and *counting* (the number of elements in regions). Any physical theory formulated on a Lorentzian manifold with a fundamental discreteness scale gives rise to a causal set by sprinkling, and the Hauptvermutung implies that this causal set essentially determines the manifold.

This applies to any framework where spacetime has a causal structure and a discrete substructure—not only causal set theory, but also lattice gauge theory (on a causal lattice), spin foam models, and other approaches to quantum gravity. The result is a statement about the *information content* of causal structure plus counting: together, they contain enough information to reconstruct the full Lorentzian geometry up to errors controlled by the density.

The scale separation condition $\lambda \gg \rho^{-1/d}$ defines the regime of validity of the manifold description. When this condition is violated—near singularities, at the Planck scale, or in regions of extreme curvature—the Hauptvermutung error diverges, signalling that the continuum approximation breaks down. The causal set remains well-defined; only its interpretation as a manifold fails. This gives a precise, quantitative criterion for where the manifold description of physics is valid and where it must be replaced by a fundamentally discrete one.

A Exponential localization of Gaussian weights

Lemma A.1 (Exponential localization). *Let $\{p_k\}$ be a Poisson process at density ρ in a d -dimensional Riemannian manifold (M, h) with curvature $|\text{Rm}| \leq 1/\lambda^2$ and $\ell \ll \lambda$. Let $W_{\text{tail}}(R) := \sum_{d(x, p_k) > R} e^{-d^2(x, p_k)/\ell^2}$ and $W_{\text{loc}} := \sum_{d(x, p_k) \leq R} e^{-d^2(x, p_k)/\ell^2}$. Then for $\alpha := R/\ell \geq 1$:*

(i) **Expected tail fraction:**

$$\frac{\mathbb{E}[W_{\text{tail}}(\alpha\ell)]}{\mathbb{E}[W_{\text{total}}]} = \frac{\Gamma(d/2, \alpha^2)}{\Gamma(d/2)} \sim \alpha^{d-2} e^{-\alpha^2} \quad \text{as } \alpha \rightarrow \infty.$$

(ii) **Tail displacement cancellation:** *The expected net displacement from the tail is even smaller. Define $V_{\text{tail}} := \sum_{d > R} w_k \exp_x^{-1}(p_k)$. By the isotropy of the Poisson process, $\mathbb{E}[V_{\text{tail}}] = 0$, and the variance is $\text{Var}(V_{\text{tail}}) = O(\rho \ell^d \cdot \alpha^d e^{-\alpha^2} \cdot \ell^2)$.*

(iii) **Probabilistic bound:** *By Markov's inequality, the probability that the tail contributes more than a fraction ε of the total weight is*

$$\mathbb{P}(W_{\text{tail}} \geq \varepsilon W_{\text{total}}) \leq \frac{\alpha^{d-2} e^{-\alpha^2}}{\varepsilon}.$$

For $\alpha = 4$ and $d = 4$: the tail fraction is $\sim 16 \cdot e^{-16} \approx 1.8 \times 10^{-6}$.

Proof. Part (i). In h -normal coordinates centred at x , the volume element is $dV_h = \sqrt{\det g} d^d x = (1 + O(r^2/\lambda^2)) d^d x$. The expected total weight is

$$\begin{aligned} \mathbb{E}[W_{\text{total}}] &= \rho \int_{\mathbb{R}^d} e^{-|x|^2/\ell^2} (1 + O(|x|^2/\lambda^2)) d^d x \\ &= \rho (\ell \sqrt{\pi})^d (1 + O(\ell^2/\lambda^2)). \end{aligned}$$

The expected tail weight uses spherical coordinates $x = r\hat{\omega}$:

$$\mathbb{E}[W_{\text{tail}}(\alpha\ell)] = \rho \Omega_{d-1} \int_{\alpha\ell}^{\infty} r^{d-1} e^{-r^2/\ell^2} dr (1 + O(\ell^2/\lambda^2)),$$

where $\Omega_{d-1} = 2\pi^{d/2}/\Gamma(d/2)$ is the solid angle. Substituting $u = r^2/\ell^2$:

$$\int_{\alpha\ell}^{\infty} r^{d-1} e^{-r^2/\ell^2} dr = \frac{\ell^d}{2} \int_{\alpha^2}^{\infty} u^{d/2-1} e^{-u} du = \frac{\ell^d}{2} \Gamma(d/2, \alpha^2).$$

The ratio $\Gamma(d/2, \alpha^2)/\Gamma(d/2)$ is the upper incomplete gamma function ratio, with the standard asymptotic $\Gamma(s, x) \sim x^{s-1}e^{-x}$ for large x .

Part (ii). By isotropy, $\mathbb{E}[w_k \exp_x^{-1}(p_k)] = 0$ for each k (the direction is uniformly distributed on S^{d-1}). The variance of V_{tail} is $\sum_{d>R} \mathbb{E}[w_k^2 |\tilde{w}_k|^2]$, which involves $\int_R^{\infty} r^{d+1} e^{-2r^2/\ell^2} dr$ —exponentially suppressed by the same mechanism.

Part (iii). Direct application of Markov’s inequality to the non-negative random variable W_{tail} . $\square$

Remark A.2 (Propagation of tail errors). The tail fraction $\sim \alpha^{d-2}e^{-\alpha^2}$ is not merely negligible—it is *superexponentially small* in α^2 , far smaller than any power of ρ or ℓ appearing elsewhere in the proof. When propagated through the derivative computation (Proposition 3.4(iv)), the tail contributes an additive error of $O(\alpha^{d-2}e^{-\alpha^2} \cdot \|D\Phi\|) = O(\alpha^{d-2}e^{-\alpha^2})$ to $D\Phi$, and similarly for the cone-preservation estimate. For any $\alpha \geq 4$, this is $\leq 10^{-6}$ in $d = 4$, utterly dominated by the statistical error $O((\rho\ell^d)^{-1/2})$ and the curvature error $O(\ell^2/\lambda^2)$.

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